🔑 **Answer Key — Teacher Copy · Lesson 7-1**

MSTAR PRACTICE WORKSHEET

Lesson 7-1: Explore Integers and Their Opposites

Grade 6 Mathematics · Standard 6.NOS.6

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Badwater Basin is 86 meters below sea level, and sea level is 0. Which integer represents that elevation?

- **A** -86
- **B** -68
- **C** 0
- **D** 86

Below sea level means less than 0, so the elevation is written with a negative sign: -86 meters.

Question 2 SELECTED RESPONSE — CORRECT: D

In Reykjavik the temperature was -9°C. What is the opposite of -9?

- **A** -18
- **B** -9
- **C** 0
- **D** 9

The opposite of a number is the same distance from 0 on the other side, so the opposite of -9 is 9.

Question 3 SELECTED RESPONSE — CORRECT: D

What is the opposite of 0?

A. A number written -0, which is less than 0

B. Every negative number

C. Zero has no opposite

D. 0

Zero is 0 units from itself, so there is no other side to move to. Zero is its own opposite.

Question 4 SELECTED RESPONSE — CORRECT: A

Which list of integers is ordered from LEAST to GREATEST?

A. -8, -3, 0, 5

B. -3, -8, 0, 5

C. 0, -3, -8, 5

D. 5, 0, -3, -8

Least to greatest reads left to right on the number line: -8, then -3, then 0, then 5.

Question 5 SELECTED RESPONSE — CORRECT: A

Badwater Basin is at -86 meters. Which elevation is FARTHER below sea level than Badwater Basin?

A. -120 meters

B. -40 meters

C. 0 meters

D. 86 meters

Farther below sea level means farther to the left of 0. -120 is 120 units below 0, which is deeper than 86 units below 0.

Question 6 SELECTED RESPONSE — CORRECT: A

Which statement about 0 is TRUE?

A 0 is an integer, and it is its own opposite

B 0 is a negative integer because it is not positive

C 0 is a positive integer because it has no negative sign

D 0 is not an integer at all

Integers are the counting numbers, their opposites, and 0. Zero is 0 units from itself, so it is its own opposite.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Badwater Basin sits 86 meters below sea level, where sea level is defined as 0. Which integer represents this elevation?

A) 0

B) −8.6

C) −86

D) 86

Below sea level means less than zero, so the elevation is −86 meters. Writing 86 would place it 86 meters ABOVE sea level.

- **A:** Zero IS sea level. The basin sits 86 meters beneath it.
- **B:** The digits are right but the size is off by a factor of ten. The depth is 86 meters.
- **D:** That places the basin 86 meters ABOVE sea level. Below zero is negative.

Part B — correct answer: C

What is the opposite of −86, and what does it represent here?

A) 0, sea level itself

B) 1/86, a fraction of the depth

C) 86, an elevation 86 meters ABOVE sea level

D) −86, the same elevation

Opposites are the same distance from zero in opposite directions. The opposite of −86 is 86, which is 86 meters above sea level.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about integers and opposites:

A) Zero is an integer.

B) The opposite of a positive integer is negative.

C) Opposites are the same distance from zero on a number line.

D) Every integer has an opposite that is a different distance from zero.

E) Zero is its own opposite.

A, B, C, and E are true. D is false — opposites are by definition the SAME distance from zero, just on opposite sides.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Describe two different real-world situations that use negative integers. For each one, tell exactly what 0 means in that situation, write one negative integer it could produce, and explain what the opposite of that integer would mean in the same situation.

Model answer: Every situation that uses negative integers has a chosen zero. With elevation, 0 is sea level, so -30 meters means 30 meters below the surface and its opposite, 30, means 30 meters above it. With temperature, 0 is zero degrees, so -5°C is five degrees below zero and its opposite, 5°C, is five degrees above. In both cases the opposite is the same distance from 0, just measured in the other direction, which is what makes it the mirror image of the original quantity.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.